TravelTide — Inspiration Review & My Plan / Inspiration & Plan

Status: 15 Oct 2026 · built from 5 past-student GitHub repos + a quick prototype check on one repo's exported raw data

Key dates / Termine

- **Wed 4 Nov** — everything finished (all deliverables)
- **Thu 5 Nov (evening) —** submission
- **Fri 6 Nov, 12:00 —** presentation practice / presentation
- Presentation format: 5 minutes to Elena (Head of Marketing). Deliverables: SQL (VS Code, Postgres URL), Python notebooks, Tableau, Streamlit app, PDF executive summary, short slides.

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1. The five past projects at a glance / Überblick

Repo	Method	Segments → perks	New perks they added	What to borrow	What to avoid
seb-bange	Weighted **perk scores** per perk (max-normalised), eligibility = top 15 %, KMeans k=6 on scores + flags, compared 4 KMeans variants + agglomerative (silhouette, DBI)	Curious Visitors (456, no-bookers) → Welcome drink · Young Explorers (846) → Exclusive discounts · Frequent Flyers → Free bag · Savvy Planners (1,910) → No cancellation fees · Premium Travelers → Free meal · Spontaneous Stayers → Free night combo	Welcome drink	Per-perk scoring idea; testing several model variants; marketing names; 1-page exec summary + 3-page report; Tableau dashboard	**52 % of users (3,092) got no perk** (within-cluster thresholds); max-normalisation is outlier-sensitive; "Savvy Planners" rely on lead time, which is almost flat in this data
krauseannelize	**Rule tree** (booking status → children → trip duration / weekday)	Lookers (479) → Exclusive discounts · Sprinters (1,767, families short) → Free meal · Voyagers (30, families long) → Free night · Trailblazers (1,557, weekday solo) → Lounge ·	Lounge access, premium upgrade	Cleanest pipeline: 3 notebooks (sessions EDA → user aggregation → segmentation), `docs/` folder with Elena's briefs, data dictionary, ER diagram, decision notes for every cleaning step; `booking_type` + `trip_category` (week/weekend/extended) features; haversine; personas in slides; segmentation-tree graphic	Two segments are tiny (30 and 59 users); **"No cancellation fee" never used**; perks outside Elena's list; no model, "validation" = bar charts only

Repo	Method	Segments → perks	New perks they added	What to borrow	What to avoid
		Weekenders (333) → Free bag · Go-Getters (1,762) → Weekday-stay discounts · Adventurers (59) → Premium upgrade			
LarsPetschke	Binary **eligibility rules** per perk (each excludes the previous), IQR outlier flags, StandardScaler + PCA + KMeans k=6, DBSCAN comparison	Free cancellation 1,191 · Free bag 2,006 · Hotel meal 648 · Discounts 162 · Flight+Hotel 26 · **No perk 1,965**	—	SQL "book" with **percentile price tiers** (Budget/Economy/Mid/Premium/Luxury at p10/34/66/90), life-stage age labels, 4-way family status, membership status from max(session_end); radar chart; age × perk heatmap; 1-pager + 3-pager	KMeans on binary flags → only 10 distinct points → silhouette 0.957 is an artefact, not "excellent clustering"; 33 % without perk; unsupported claims ("+20–30 % CTR")
SurayaSumona	One **SQL query**: cohort CTE → metrics CTE → CASE waterfall → perk; Tableau demographics	Groups 22.7 % → No cancellation fees · Family 21.2 % → Exclusive discounts · Couple 15.7 % → Free bag · Frequent Traveler 14.8 % → Priority boarding · Business 9.2 % → Free night · Solo 8.0 % → Free hotel meal · Other 8.4 % → 10 % first-	Priority boarding, 10 % first-booking discount	Whole segmentation readable in one SQL file; demographic perk analysis (age, gender, marital, children) in Tableau; recommendations on tiers/points	Waterfall order decides everything but is not justified; text and code disagree (Couple "2 seats" not in code); `DATE_PART('minute', …)` only returns the minutes component, not the duration

Repo	Method	Segments → perks	New perks they added	What to borrow	What to avoid
		booking discount			
svechino	Six SQL **indexes** (min-max scaled) → rank users per index → perk = best relative rank; KMeans k=5 for comparison	Free bag 1,096 · Free meal 955 · Exclusive discount 905 · Free night w/ flight 807 · No cancellation fee 758 · **Active Searchers 1,477**	—	**Bargain Hunter Index** = discount share × avg discount × scaled *dollars saved per km* (matches Elena's exclusive-discount brief); Hotel Hunter Index (dollars saved per night); rank-based assignment → everyone gets a perk; haversine compared 3 ways	KMeans mixed in demographic booleans (they dominate distance); `DATE_PART('second', …)` bug; copy-paste leftovers ("Metrocar") in the report

Common ground: all use Elena's cohort (sessions after 2023-01-04, users with > 7 sessions → ~5,998 users, ~49k sessions). All map segments to Elena's five perks; three of them also added perks of their own.

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2. Data facts & traps / Datenfakten & Fallen

From the repos plus my own quick check on krauseannelize's raw extract (49,211 sessions, 5,998 users). Recompute all numbers with your own pipeline.

- **Reference date:** last session ends **2023-07-28** → use it for age and tenure (not today's date — some repos inflated ages by 1.5+ years).
- **Cancelled trips:** a cancellation session repeats the original `trip_id`. Count a trip as *valid* only if that trip_id has no cancellation session. Check result: ~15,489 valid trips, ~610 cancelled trip_ids, **~556 users with zero valid trips**.
- **Hotel nights:** some negative / zero / inconsistent → recompute from check-in/check-out dates.
- **Durations in SQL:** use `EXTRACT(EPOCH FROM session_end - session_start)/60`, not `DATE_PART`.
- **Flat signals (weak for segmentation):** sessions per user (median 8), clicks per minute (~8 for everyone → clicks and duration are the same signal), booking lead time (median ~7 days, 90 % ≤ 10 days).
- **Strong signals:** seats, rooms, checked bags, nights, trip distance (haversine home airport → destination), weekend vs weekday start, discount share, hotel-only vs flight+hotel, repeat destinations.
- **Who:** ~88 % female; ~83 % USA / ~17 % Canada; flight+hotel bookings dominate (~72 % of a user's trips on average).
- **Modelling lessons:** don't cluster on binary flags (fake silhouette) or demographics (they dominate); don't leave users without a perk; keep every segment ≥ 4–5 % of users.

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3. My approach / Mein Ansatz (hybrid, fits the decision already made)

Theme: **"Remove the one friction that stops each traveller booking the next trip."** Every perk removes a friction (price, risk, luggage, hotel costs, commitment).

1. **SQL (VS Code):** cohort → clean session-level table with trip flags.

2. **Python:** user-level features (~25) → **rule-based segments** in a documented priority order → everyone else gets the perk with the highest **perk-affinity score** (percentile-ranked, svechino-style ranking but robust).
3. **Models as evidence:** KMeans on continuous behaviour features to check / refine the rules; a **decision tree** trained to reproduce the segments → proves the rules are consistent and lets TravelTide score all ~1M users later (business reason for the model).
4. **Perks in two layers:** a **core perk** from Elena's five + a **pilot perk** (new idea) per segment, to be A/B tested.

Proposed segments (prototype sizes on 5,998 users, priority order as listed)

#	Segment	Rule sketch	Size	Core perk (Elena's 5)	Pilot perk (new)
1	**Dreamers**	0 valid trips, still browsing	556 (9 %)	No cancellation fees ("book without risk")	★ **48-h Fare Lock** — hold today's price for free
2	**Family Crew**	has children AND avg seats or rooms ⩾ 1.5	417 (7 %)	Free hotel meal ("kids eat free")	Family fare — kids' seats discounted
3	**Friends & Groups**	no children AND seats or rooms ⩾ 2	362 (6 %)	No cancellation fees	Organiser reward — 5th seat/room at half price
4	**Weekday Commuters**	no Fri/Sat starts, ⩽ 3 nights, 1 seat, almost no bags	301 (5 %)	No cancellation fees	Lounge / fast-track pass
5	**Weekend Escapers**	⩾ 50 % Fri/Sat starts, ⩽ 3 nights	598 (10 %)	1 night free hotel with flight ("make it a long weekend")	Late Sunday check-out
6	**Favourite-City Returners ★**	same destination ⩾ 2 times	491 (8 %)	1 night free hotel with flight	★ City stamp card — 3rd trip to your city earns a local extra
7	**Deal Hunters**	high discount share (refine with Bargain Hunter Index)	449 (7.5 %)	Exclusive discounts (Elena's brief)	Price-drop guarantee
8	**Long-haul Packers**	avg bags ⩾ 1 OR avg distance ⩾ 3,000 km	1,091 (18 %)	Free checked bag	Extra-legroom seat on long-haul
9	**Solo Stayers**	⩾ 50 % hotel-only trips (avg 7.5 nights, ~700 km from home)	297 (5 %)	Free hotel meal	Solo welcome dinner (first evening)
—	**All-rounders**	everyone else	1,436 (24 %)	Highest perk-affinity score	—

Prototype observations worth checking in your own data: Favourite-City Returners book the most (avg ~4.3 trips vs ~2.6) — partly mechanical (more trips → more repeats), say so; Solo Stayers look like long hotel stays close to home; Weekday Commuters have almost no bags (avg 0.06).

Signature picks (yours, not in any repo): segment = **Favourite-City Returners**; perk = **48-h Fare Lock** for Dreamers. Alternatives: segment "Long-stay Staycationers" (= Solo Stayers reframed); perks "Late Sunday check-out", "Price-drop guarantee".

Fewer segments if needed: merge Friends & Groups into Family Crew ("Party Travellers") and Weekday Commuters into Solo Stayers → 7 + All-rounders.

4. Features to build / Features (user level)

- **Activity:** sessions, avg session minutes (choose either minutes or clicks), valid trips, flights, hotels, cancellations, cancellation rate
- **Trip mix:** share flight+hotel, share flight-only, share hotel-only, return-flight share
- **Party:** avg seats, avg rooms, avg checked bags, bags per seat
- **Length & distance:** avg nights (recomputed), avg trip days, avg distance km, share long-haul (> 3,000 km)
- **Timing:** share Fri/Sat starts, share Mon–Thu starts, avg lead days (expected flat — show it once)
- **Price:** discount share, avg discount %, dollars saved per km (Bargain Hunter Index), dollars saved per night (hotel), total spend after discount, spend per trip, price tier (p10/34/66/90)
- **Loyalty:** repeat-destination count, distinct destinations
- **Demographics:** age (ref 2023-07-28), tenure months, married, has_children, family status (4-way), home country
- **Perk-affinity scores (5):** percentile ranks, one per Elena perk

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5. Models for the "how we got here" story

- **KMeans:** 8–10 continuous behaviour features, log-transform skewed ones, standardise; k from elbow + silhouette (expect modest scores ~0.15–0.35 — say why that's normal for behaviour data); PCA 2-D plot; crosstab clusters vs rule segments + Adjusted Rand Index; use what differs to tune thresholds.
- **Decision tree (depth 4–5):** features → final segment; accuracy, tree plot, feature importance; saved with joblib; powers the Streamlit "Perk Finder".

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6. Deliverables plan / Abgaben

Repo: `sql/01_explore.sql, 02_cohort_sessions.sql, 03_user_features.sql` · `notebooks/01_eda_cleaning, 02_features, 03_segments_models` · `data/` (small exports) · `app/` (Streamlit) · `reports/` (PDF, slides) · `docs/decision_log.md, data_dictionary.md` · README with story, links and one line crediting inspirations. Keep the DB URL in `.env` / `st.secrets`, never in the repo.

Notebook visuals: data-quality before/after table, key distributions, correlation heatmap, **rule-overlap heatmap** (justifies the priority order), segment sizes, segment profile heatmap or radar (z-scores), elbow + silhouette, PCA scatter, clusters-vs-rules heatmap, decision tree.

Tableau story (3 dashboards): (1) Who our customers are — age, family status, country, gender; (2) How they travel — trip types, weekday × trip-length heatmap, destination map from home airports, nights vs distance, discounts; (3) Segments & perks — sizes, profiles, perk map, perk cost tier. Publish on Tableau Public.

Streamlit pages: Story (headline KPIs) · Data & cohort (funnel 1.02M users → 5,998) · Segments (tree, sizes, profiles) · Model check (KMeans + tree) · **Perk Finder** (pick a user or move sliders → segment + core + pilot perk) · Tableau (embedded) · Measuring success. Data from CSV, no live DB.

PDF executive summary (1 page + optional appendix): objective, approach, 3 key findings, segment–perk table, recommendation, how to measure.

Slides (5 min, ~7 slides, ~40 s each): hook number (e.g. share of Dreamers) → cohort & approach → segments at a glance → signature segment + perk spotlight → perk map (core + pilot, cost tier) → how we'll measure → recommendation.

Measuring success (A/B): per segment 10 % holdout (no perk), split the rest core vs pilot; primary KPI booking conversion within 60 days of the invite; secondary: rewards sign-up rate, email CTR, bookings per user, revenue per user, cancellation rate, perk cost per extra booking; 8–12 weeks; refresh segments quarterly; scale to all users with the decision tree.

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7. Roadmap / Fahrplan

Weekend sprint / Wochenend-Sprint (Sat 17 – Sun 18 Oct) — finish Week 1

Saturday

1. Setup (1 h): GitHub repo + folders, VS Code Postgres connection, `.env`, Python env (pandas, sqlalchemy, psycopg2, scikit-learn, seaborn, plotly, streamlit).
2. Schema exploration (1.5 h): row counts, grain, keys, fact vs dimension, ER diagram (dbdiagram.io or draw.io) → `sql/01`.
3. Cohort + session table (2 h) → `sql/02`: filters, joins, valid/cancelled trip flags, recomputed nights, trip start/end, start weekday, lead days, booking type, discount dollars, distance.
4. Sanity check in notebook (1 h): ~49k sessions, ~6k users, ~15.5k valid trips, ~556 zero-trip users.

Sunday

1. Notebook 01 EDA (2 h): data-quality table (issue → evidence → decision), key distributions, note flat vs strong signals.
2. User-level table (2 h) with the features in section 4 + `data_dictionary.md`.
3. Decision log started (1 h); confirm signature segment & perk; sketch the rule tree on paper.
4. Push to GitHub; list open questions.

Week 2 (Mon 19 – Sun 25 Oct) — features & first segments

- Mon–Tue: perk-affinity scores, Bargain Hunter Index, cancellation metrics
- Wed: segment rules v1, overlap heatmap, sizes, threshold tuning (≥ 4–5 % each)
- Thu: segment profiles + one real example customer per segment (personas)
- Fri: Tableau dashboards 1 and 2
- Weekend: buffer + Streamlit skeleton (multipage, data load, Tableau embed test)

Week 3 (Mon 26 Oct – Sun 1 Nov) — models & story

- Mon: KMeans check · Tue: decision tree + saved model · Wed: final perk mapping, pilot perks, cost tiers, A/B plan
- Thu: Tableau dashboard 3 + Story, publish · Fri: fill Streamlit pages
- Weekend: deploy Streamlit (Community Cloud), buffer

Week 4 (Mon 2 – Wed 4 Nov) — communicate

- Mon 2: slides + speaker notes, first timed run
- Tue 3: PDF executive summary, README, notebooks "restart & run all"
- **Wed 4: final QA —** same numbers in notebooks, Tableau, Streamlit, PDF, slides; all links work → **done**
- Thu 5: 2–3 timed rehearsals, offline backup (PDF slides, app screenshots), **submit in the evening**
- **Fri 6, 12:00:** presentation practice / presentation

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8. Make it yours / Eigene Handschrift

- Own naming theme and one colour palette across Tableau, Streamlit and slides.
- Decision log in your own words, including dead ends (e.g. "lead time looked useful but is flat").
- Signature segment + signature perk; personas from real user_ids with their numbers.
- Write SQL and code yourself; credit inspirations in one README line (check Masterschool's rules).
- Only claims you can back with your data — no invented uplift percentages.

9. Open decisions / Offene Entscheidungen

1. Keep 9 segments or merge to 7?
2. Signature pair OK (Favourite-City Returners + 48-h Fare Lock)?
3. Dreamers' core perk: no cancellation fees (my pick) or exclusive discount (what most repos did)?
4. Distance in SQL (function) or Python?
5. Public deployment of Streamlit / GitHub repo?

Sources: github.com/seb-bange/travel-tide_mastery-project · github.com/krauseannelize/report-traveltide-segmentation · github.com/LarsPetschke/Mastery_Project_TravelTide · github.com/SurayaSumona/Travel-Tide-Mastery-Project · github.com/svechino/TravelTide